

System Architecture of the AMD MI400 Series GPU

Steve Scott

Corporate Fellow, Network & System Architecture

David Riddoch

Senior Fellow, Networking

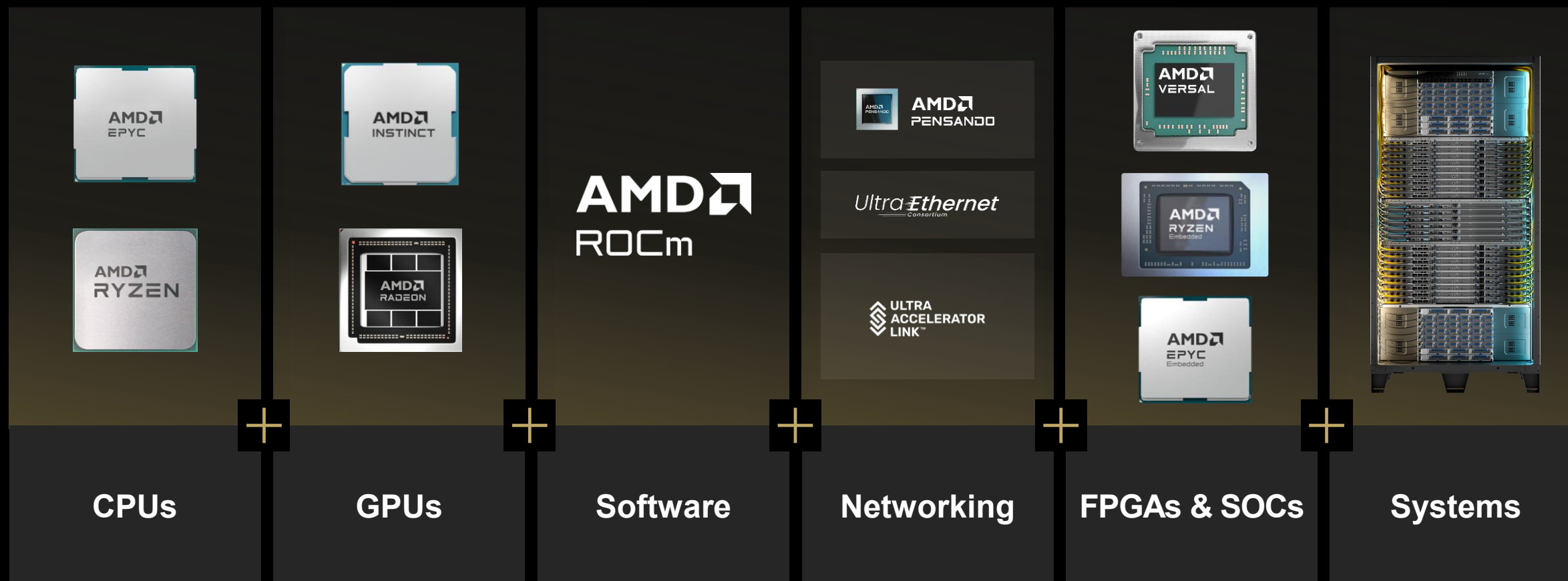
Krishna Doddapaneni

Corporate Vice President, Networking

Hot Chips



AMD Focused on Full System Solutions



Co-Designed System Architecture for Rack-Scale AI

Built from the Highest Performing CPU, GPU, and AI-NIC

EPYC “Venice”

6th Gen EPYC CPU



96

High Frequency Cores

Gen6

PCIe CPU to GPU

1.6 TB/s

Memory Bandwidth

Instinct MI455

5th Gen Instinct GPU



40 PF

FP4 Flops

432 GB

HBM4 Capacity

23.3 TB/s

Memory Bandwidth

Pensando “Vulcano”

3rd Gen Pensando AI-NIC



800 Gbps

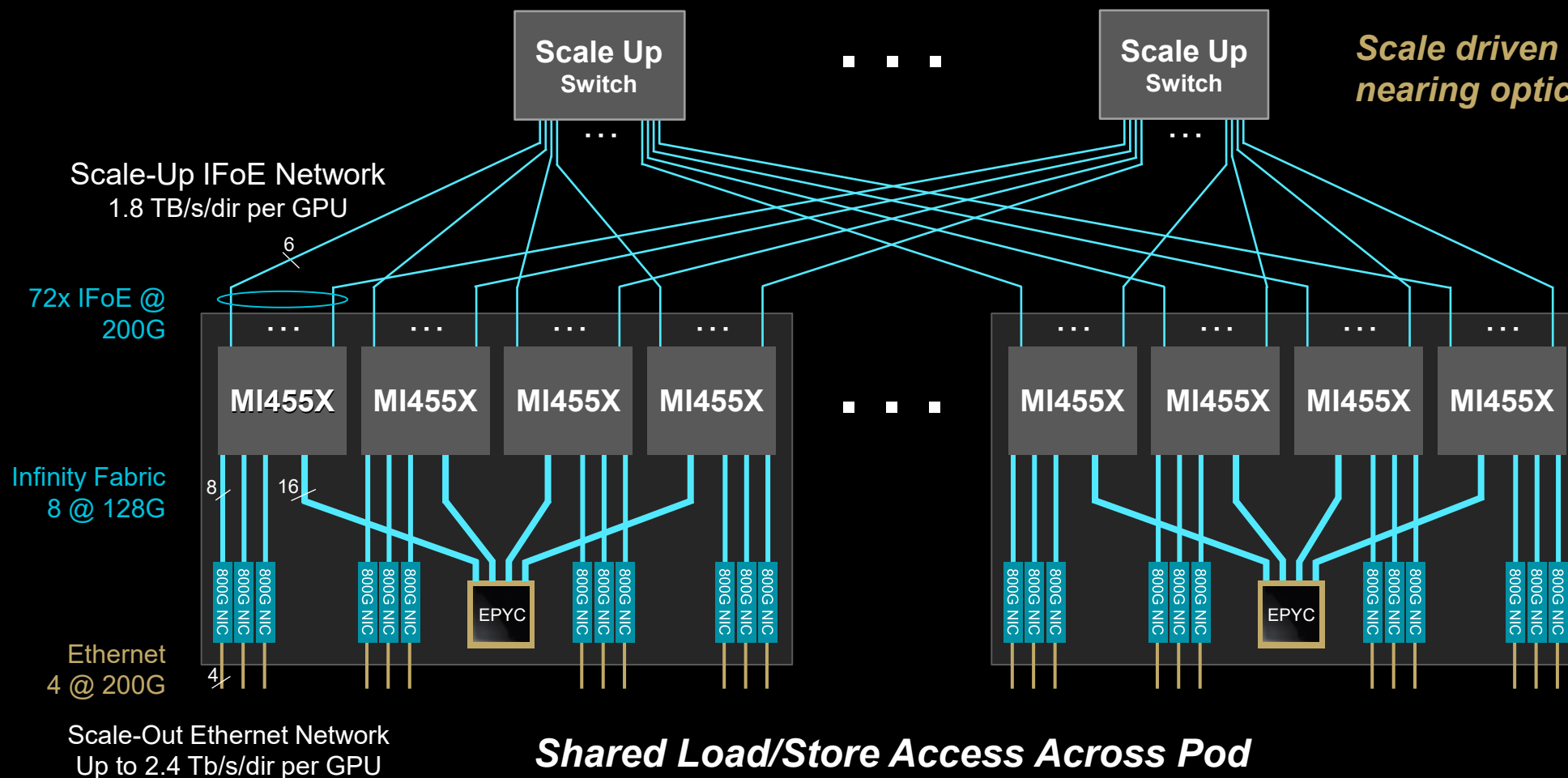
Bandwidth

200M

Packets per sec

Fully Programmable
Protocol Support

**Scale driven by electrical signaling;
nearing optical transition**



AMD Helios™ Highlights

The Rackscale AI Infrastructure

72 GPUs

Per Rack

31 TB

HBM4 Memory

260 TB/s

Scale Up Bandwidth

2.9 Exaflops

AI Compute

1.7 PB/s

HBM4 Bandwidth

43 TB/s

Scale Out Bandwidth



ESUN



AMD Helios Rack Construction

Management & Power

TOR Switch + RMC
Management SW

Compute

18x 10U Compute Trays

Switching

6x 10U Switch Trays

Power

Redundant power supplies
50vDC LC busbar

Compute Trays
Switch Trays
Compute Trays



Specifications

44OU, ORW-HPR

72 GPU Config

Rear BlindMate QD liquid-cooling

Scale-Up Fabric

4x Cable Cartridges

Leak Detection

The Building Blocks of Helios

Compute Tray – Combining compute, host, memory and networking

GPU Compute & Scale Up

4x AMD Instinct MI455X EAMs
12x 3x2 UALoE Links per EAM

CPU Host

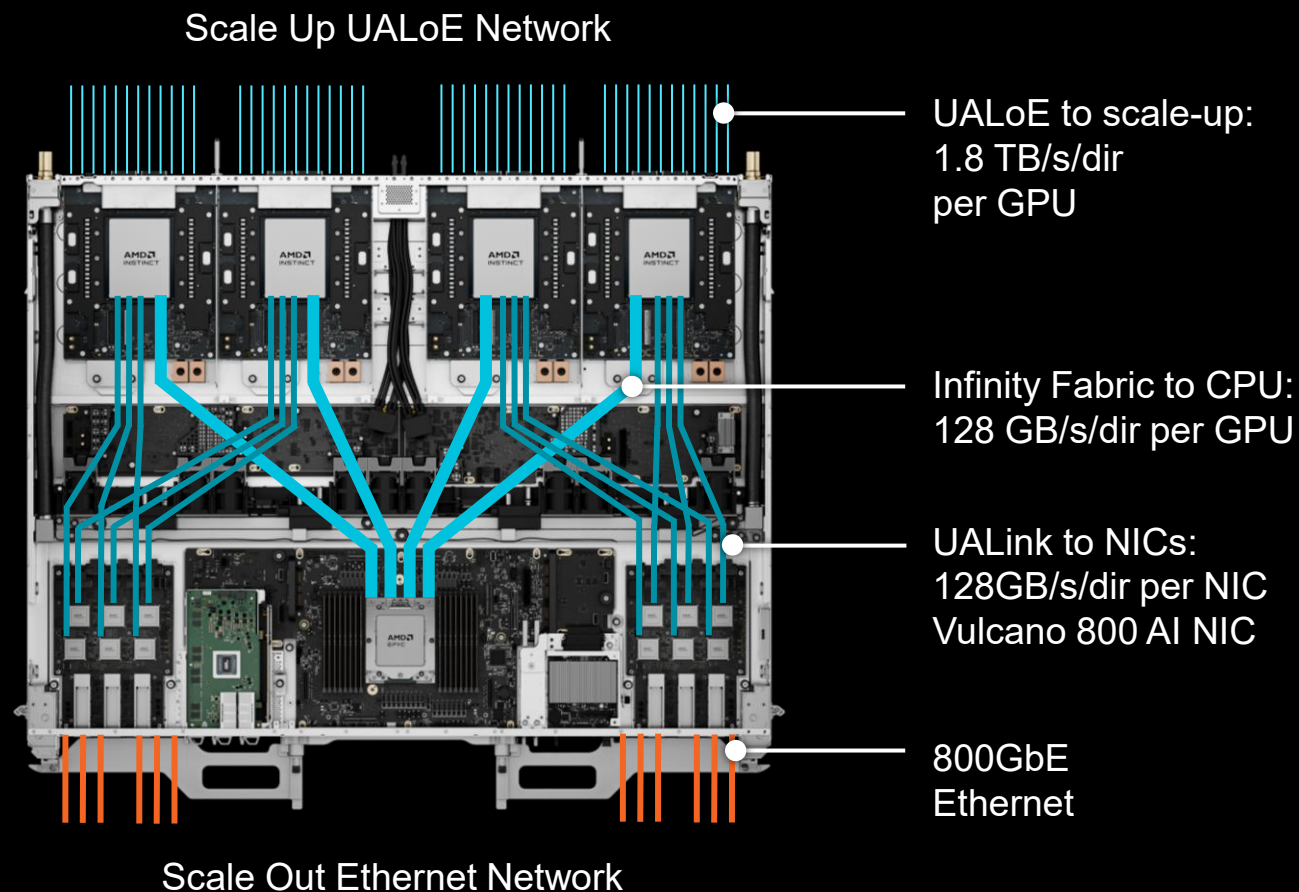
1x AMD “Venice” SP7 CPU
Coherent Infinity Fabric to MI455X

Scale Out Network

Up to 3x “Vulcano” AI NIC Per EAM
UALink (x8) to MI455X EAM

Front-End Network

1x FH PCIe CEM NIC



The Building Blocks of Helios

Switch Tray – Combining compute, memory and networking

Scale Up Fabric

2x 512 port 200G UALoE Switch ASICs
72x 3x2 UALoE active links per ASIC

High-Speed Connectivity

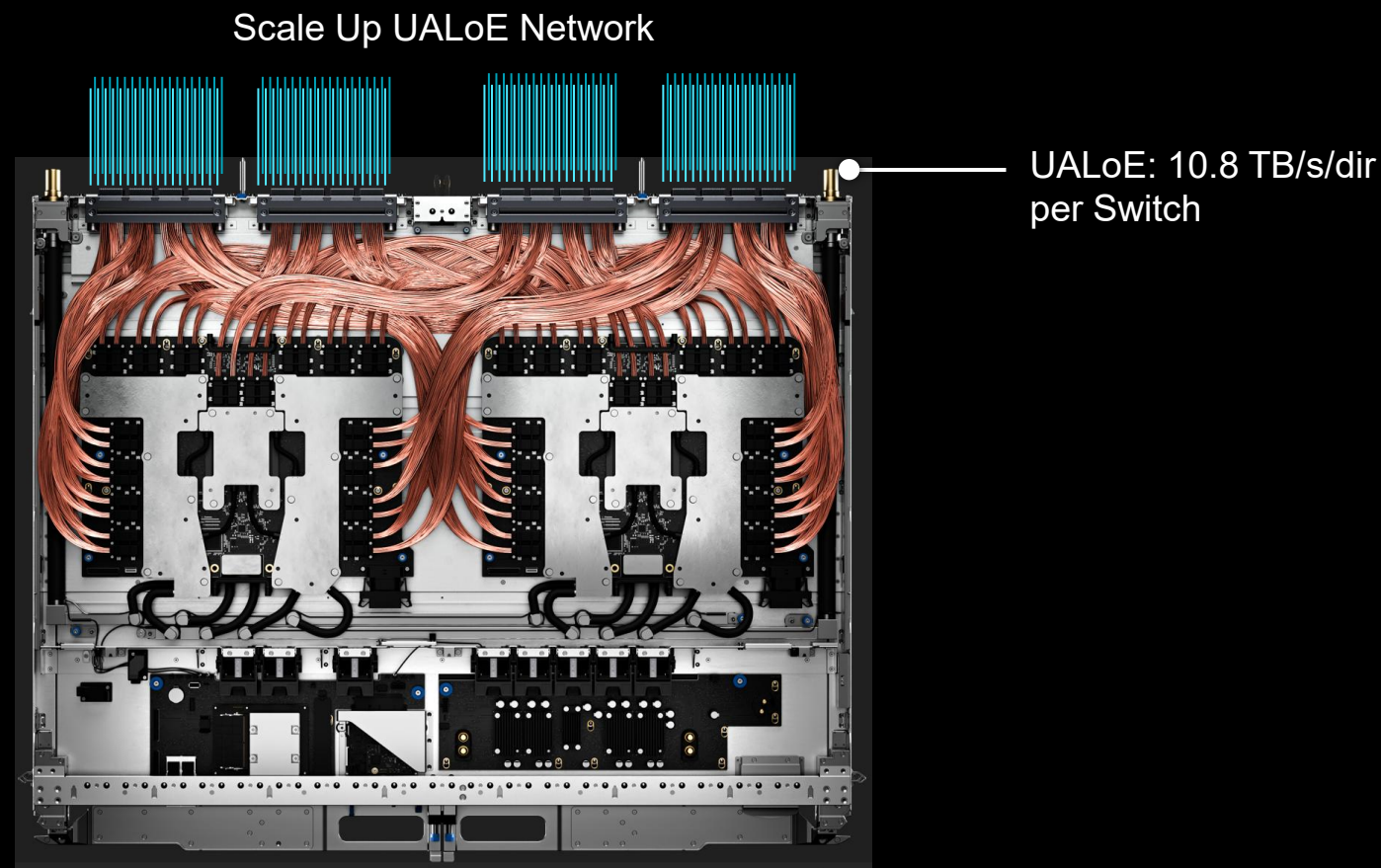
1x High-BW Copper Scale-up

Fabric Architecture

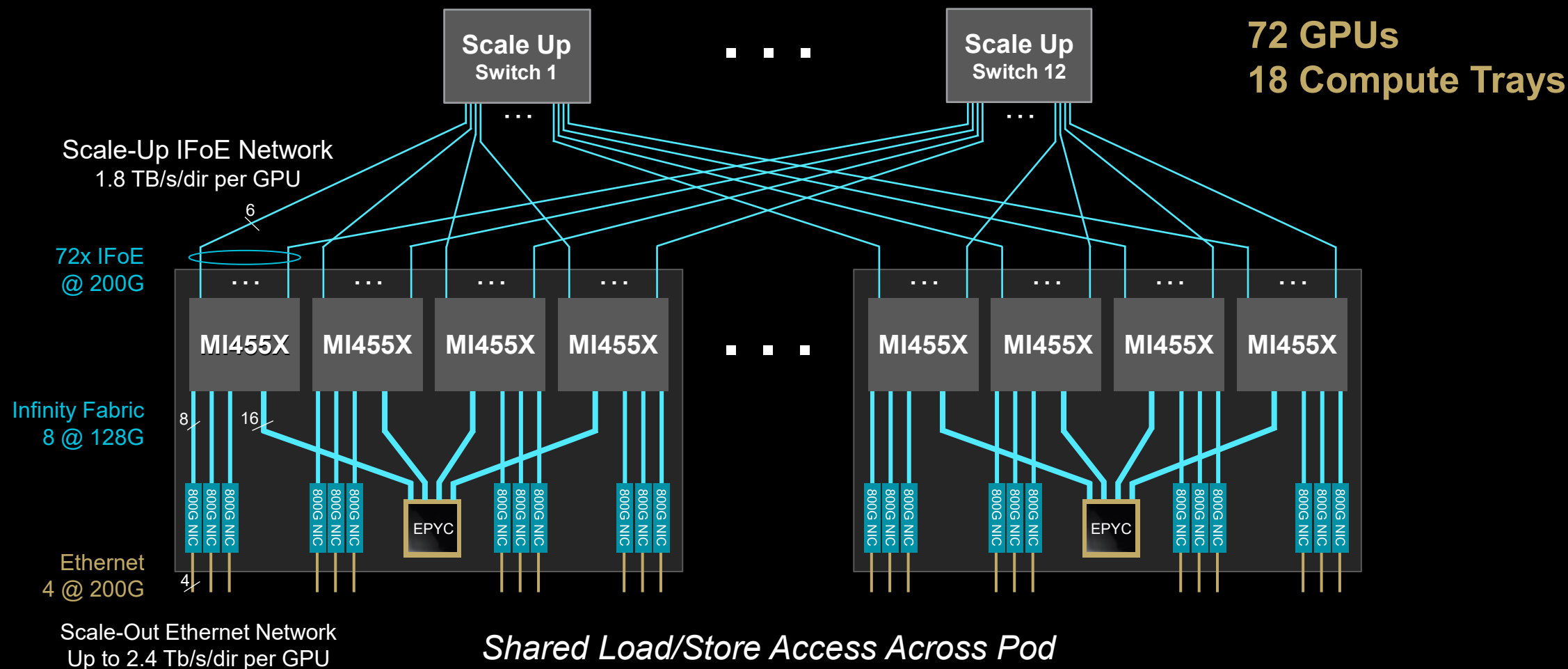
UALoE Multi-Plane Network
All To All GPU Connectivity

Management and Power

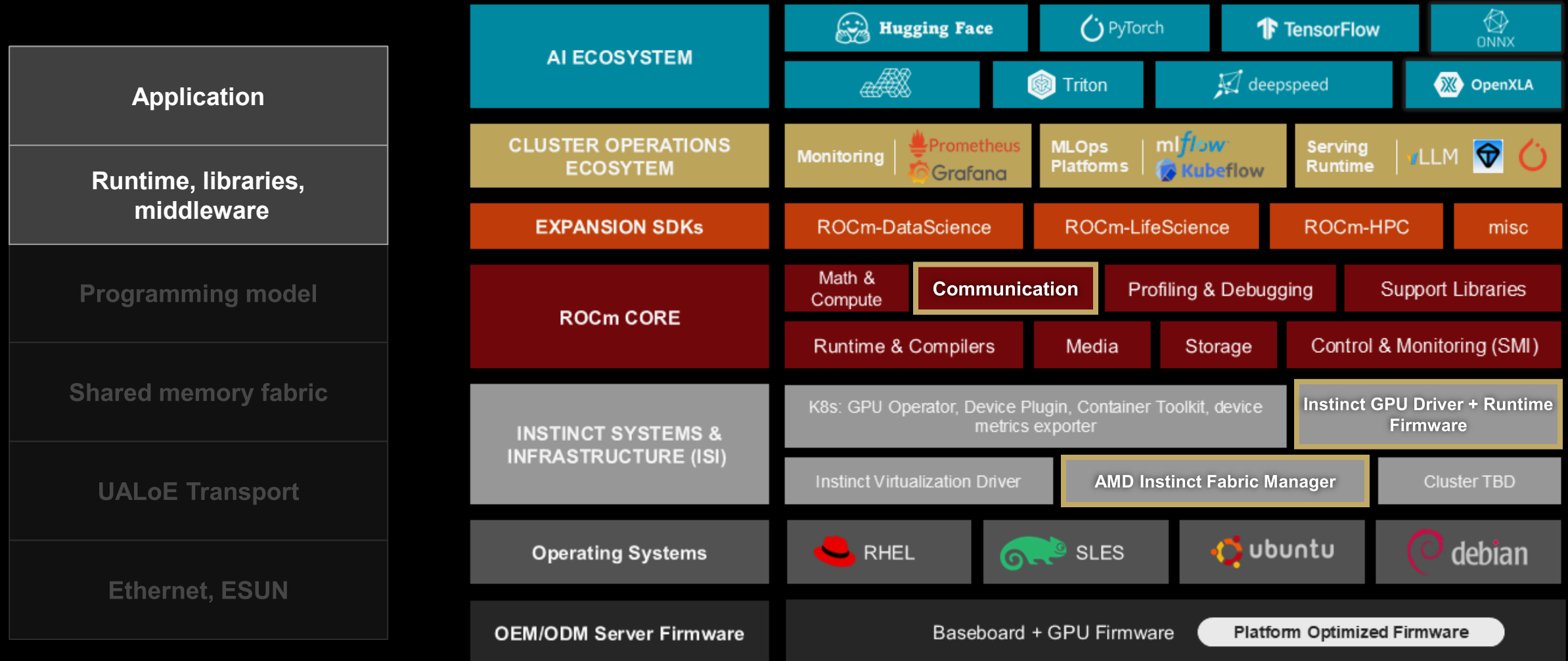
AMD Ryzen + Run BMC
Liquid Cooled ~7kW Infrastructure



Scale-Up Pod Architecture in Helios

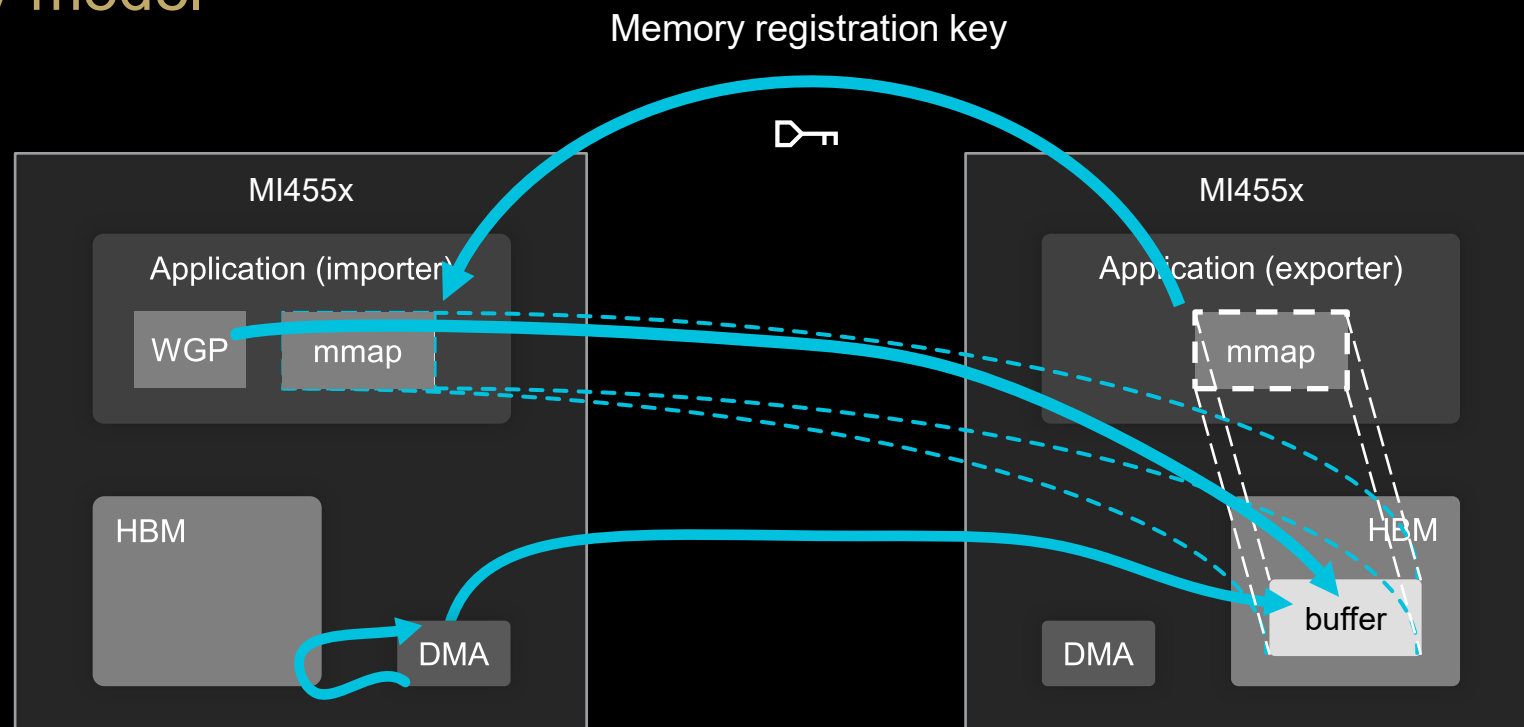
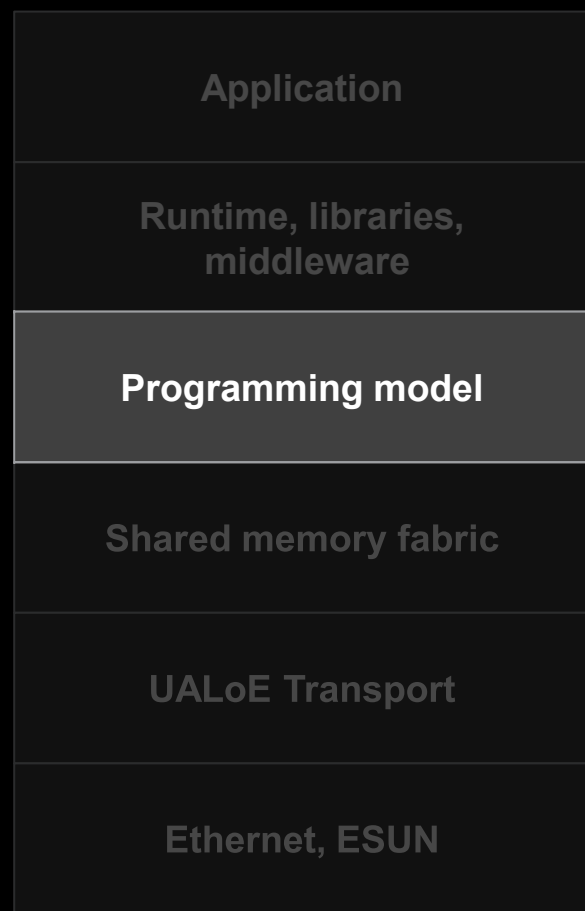


UALoE Technology



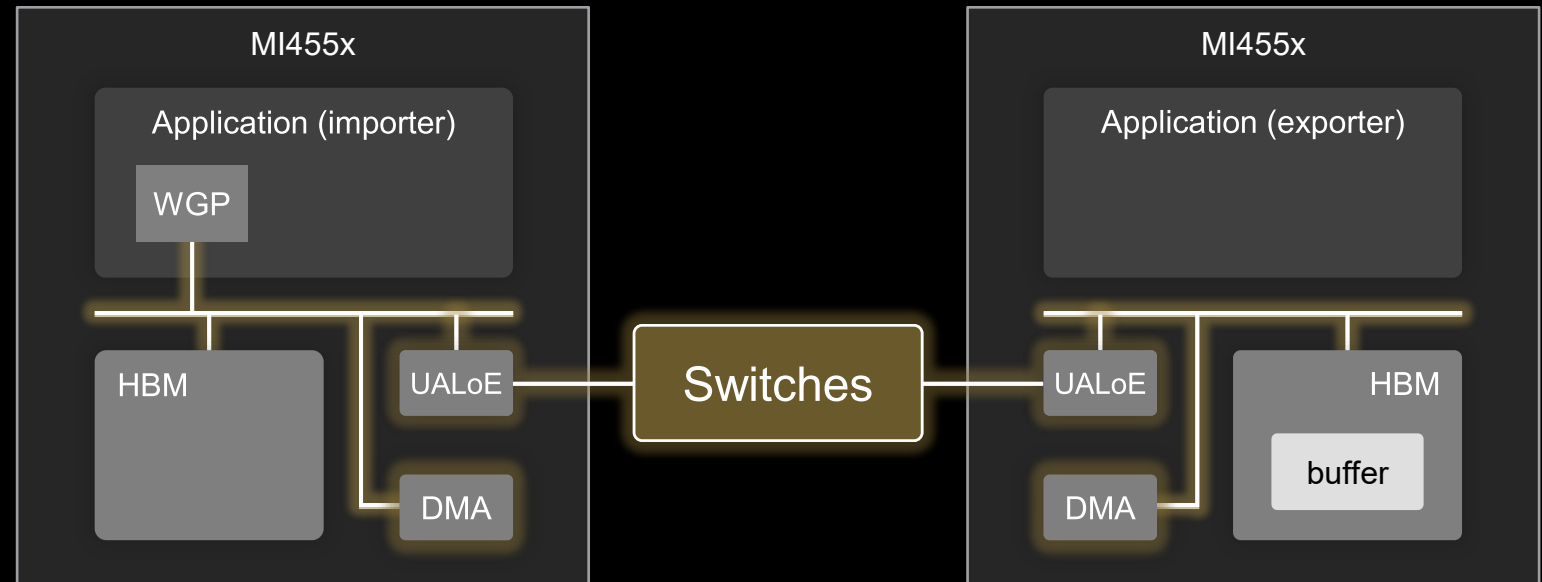
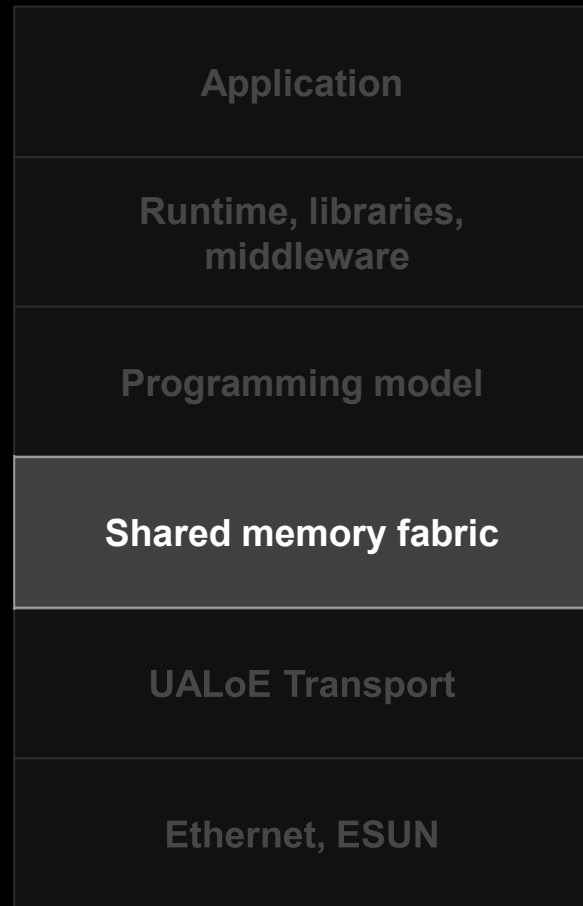
UALoE Technology

Distributed shared memory model



UALoE Technology

Load/store fabric spanning the whole Helios rack

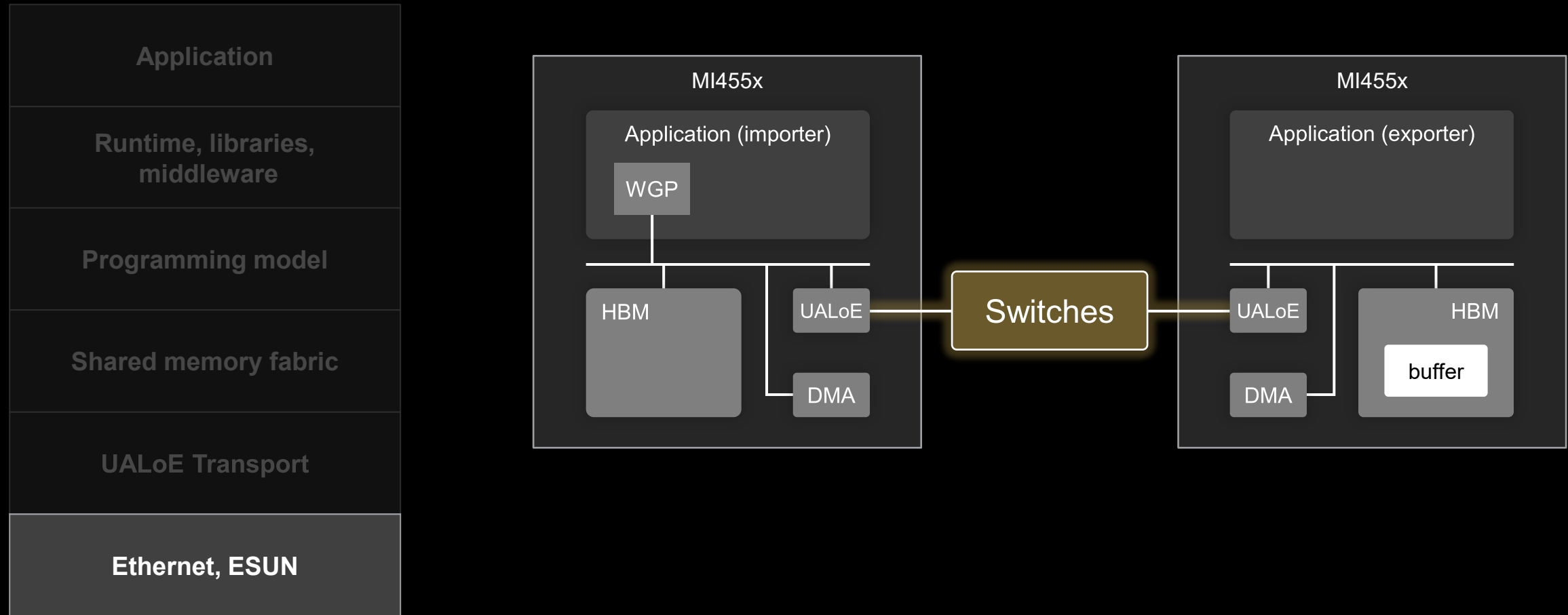


Topology-aware DMA engines

Offload data movement from WGPs in parallel with compute
Frontend / backend architecture distributes load dynamically
Backends are adjacent to UALoE links, optimizing data movement

UALoE Technology

Built on open Ethernet and ESUN standards and Broadcom switches

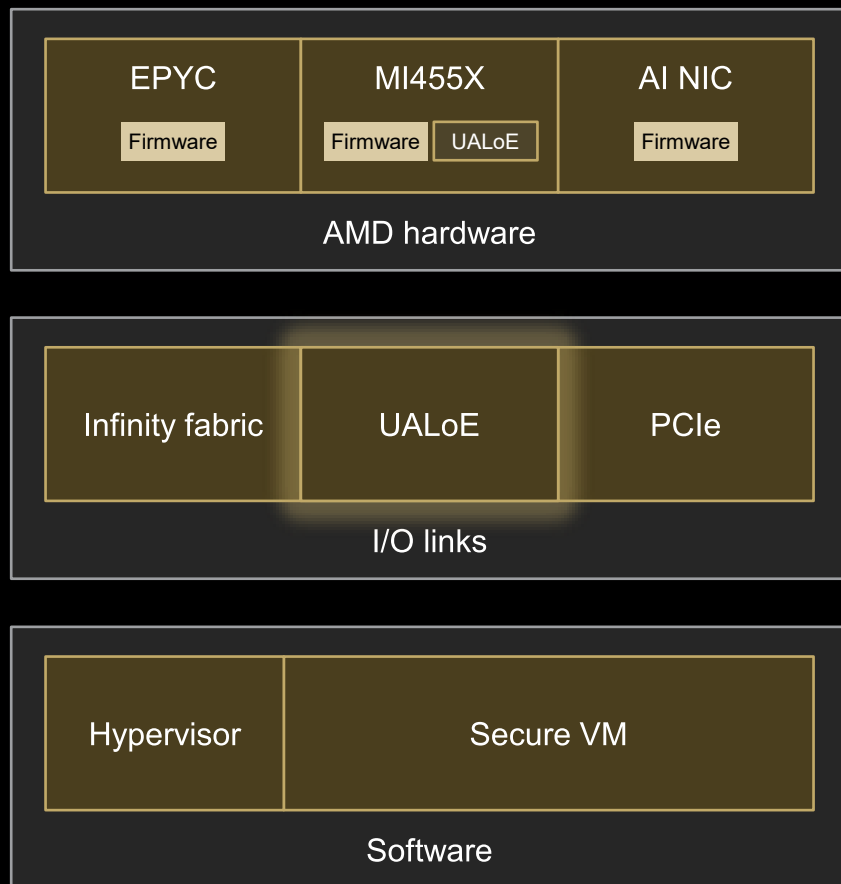


UALoE Technology

Application	UALoE integrated network adapters - Each MI455X integrates 18x 800 Gbps UALoE adapters - Fabric interface, transport, Ethernet, security, management - Requires a highly optimized design
Runtime, libraries, middleware	
Programming model	UALoE transport protocol - A lightweight, reliable protocol designed for scale-up - Minimal overheads for maximum bandwidth efficiency - Dynamic packing to optimize for bandwidth vs. latency - Leverages Ethernet PFC flow control
Shared memory fabric	
UALoE Transport	UALoE reliability and resilience - Packet loss recovered via link-layer replay or end-to-end retransmission - Failure of link or switch: Retransmission via an alternative network plane - After transient failure (link flap) or repair, network plane can be brought back online to restore full performance
Ethernet, ESUN	

Security

Rack-scale Confidential Computing¹



AMD hardware

Secure by design features
Virtualisation, partitioning
External memory encryption (SEV-SNP)

Secure boot

Root of trust: DICE identity and attestation

Universal link encryption

GPU↔GPU, CPU↔GPU, GPU↔AINIC
Confidentiality and integrity protection

Trusted execution environment

TDISP protects VM↔device interfaces

UALoE security

End-to-end packet
encryption

Switch is outside
of TCB

Fast key rotation

Periodic re-keying of
master keys

Virtual pods for flexible partitioning

Virtual Pods (VPods)

Flexible

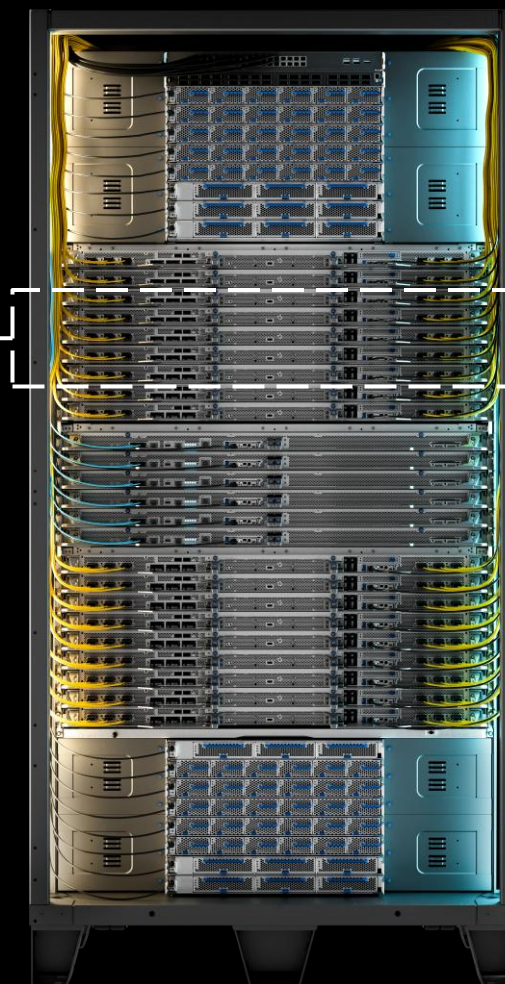
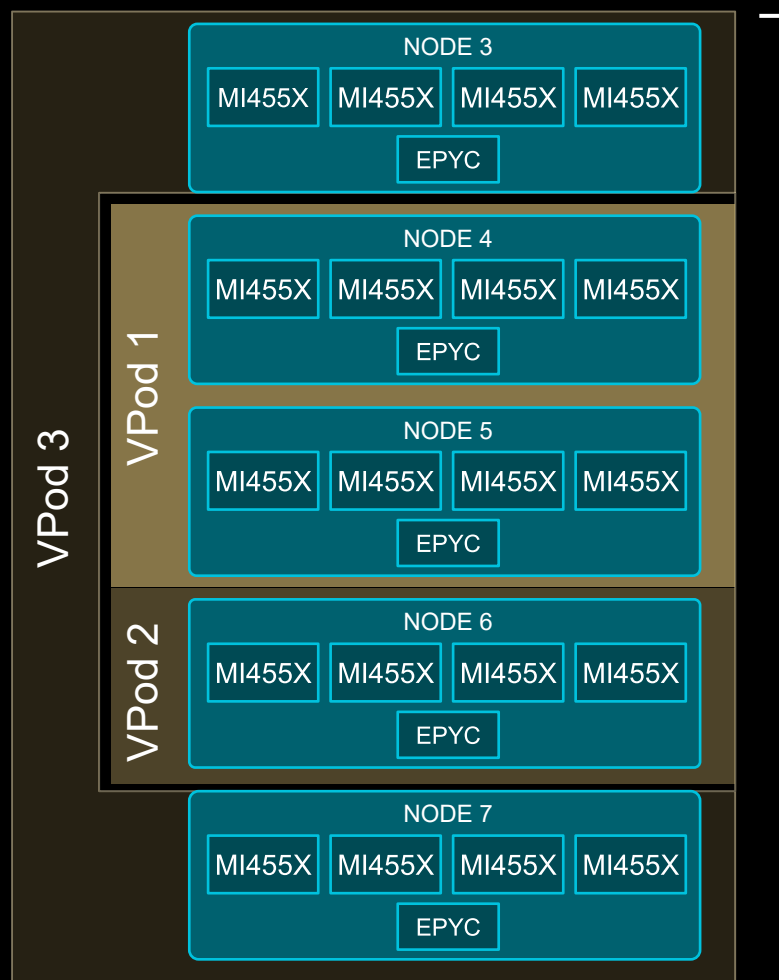
Combine subset of nodes to form a Virtual Pod

Secure

VPods are isolated from each other

Resilience

Node level failures are contained to a VPod



Scale-up fault resilience

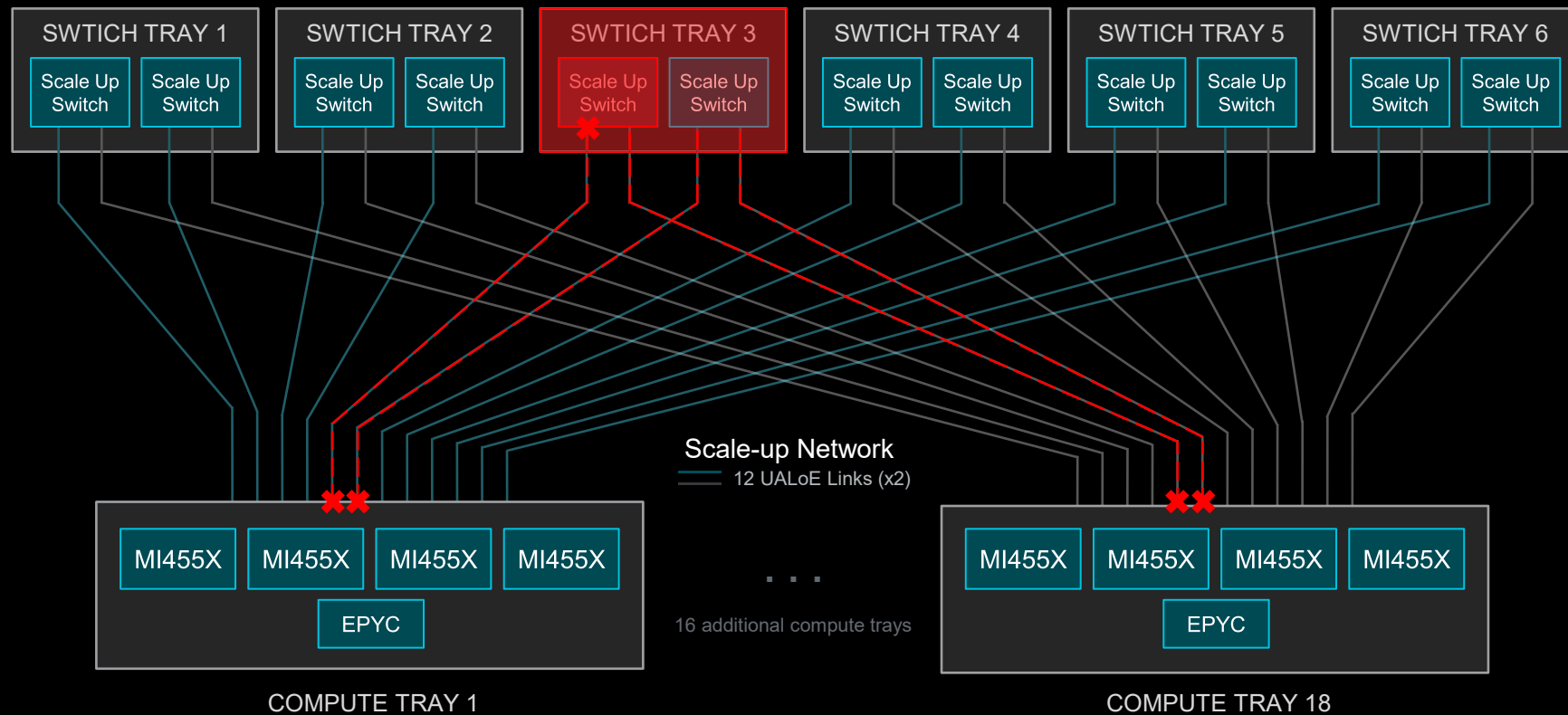
NORMAL OPERATION

SINGLE LINK FAILURE

SINGLE SWITCH FAILIRE

SWITCH TRAY FAILURE

Soft errors recovered via reliable link layer in hardware



Scale-up fault resilience

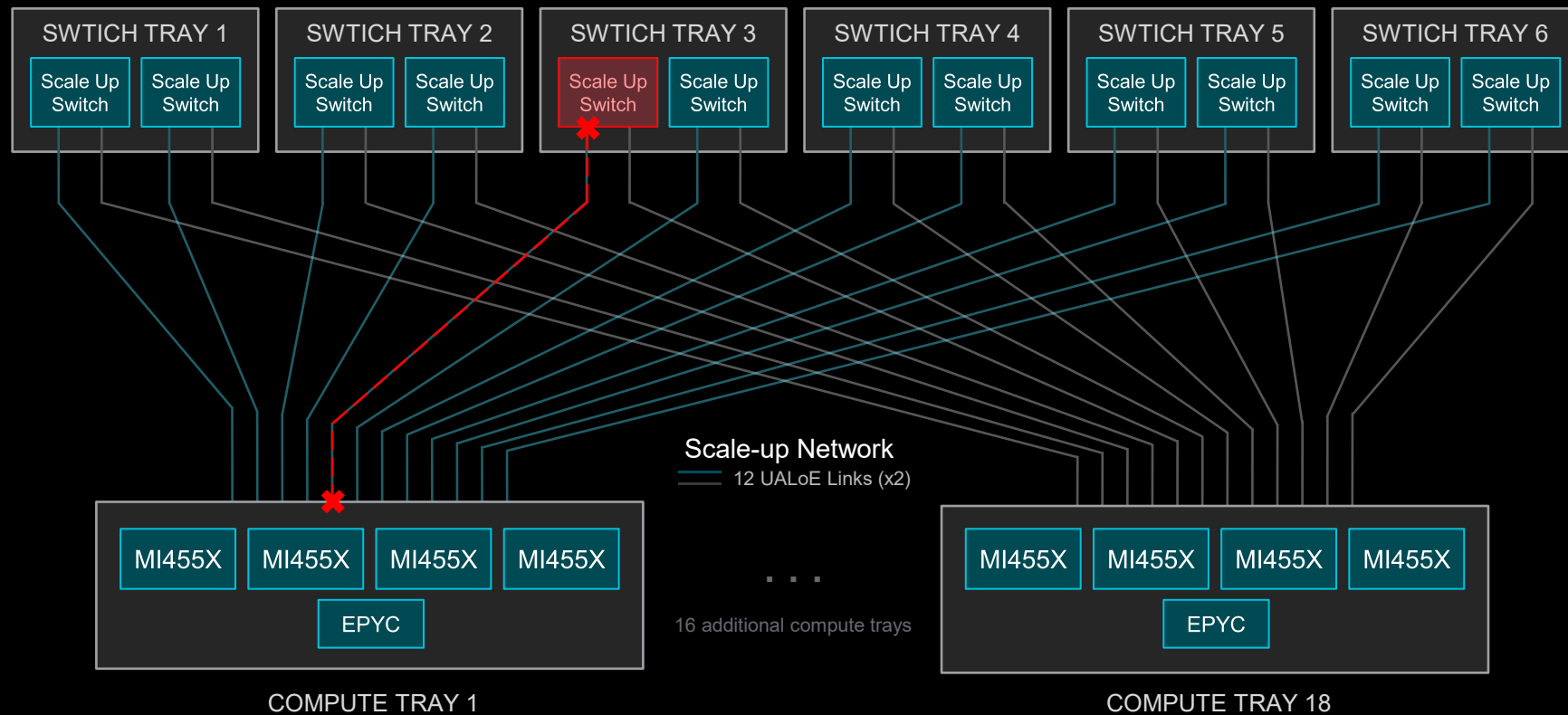
NORMAL OPERATION

SINGLE LINK FAILURE

SINGLE SWITCH FAILIRE

SWITCH TRAY FAILURE

Single link lost | Impacts local VPod only
DMA rebalances over all remaining links | WGP refs rebalance under SW control



Scale-up fault resilience

NORMAL OPERATION

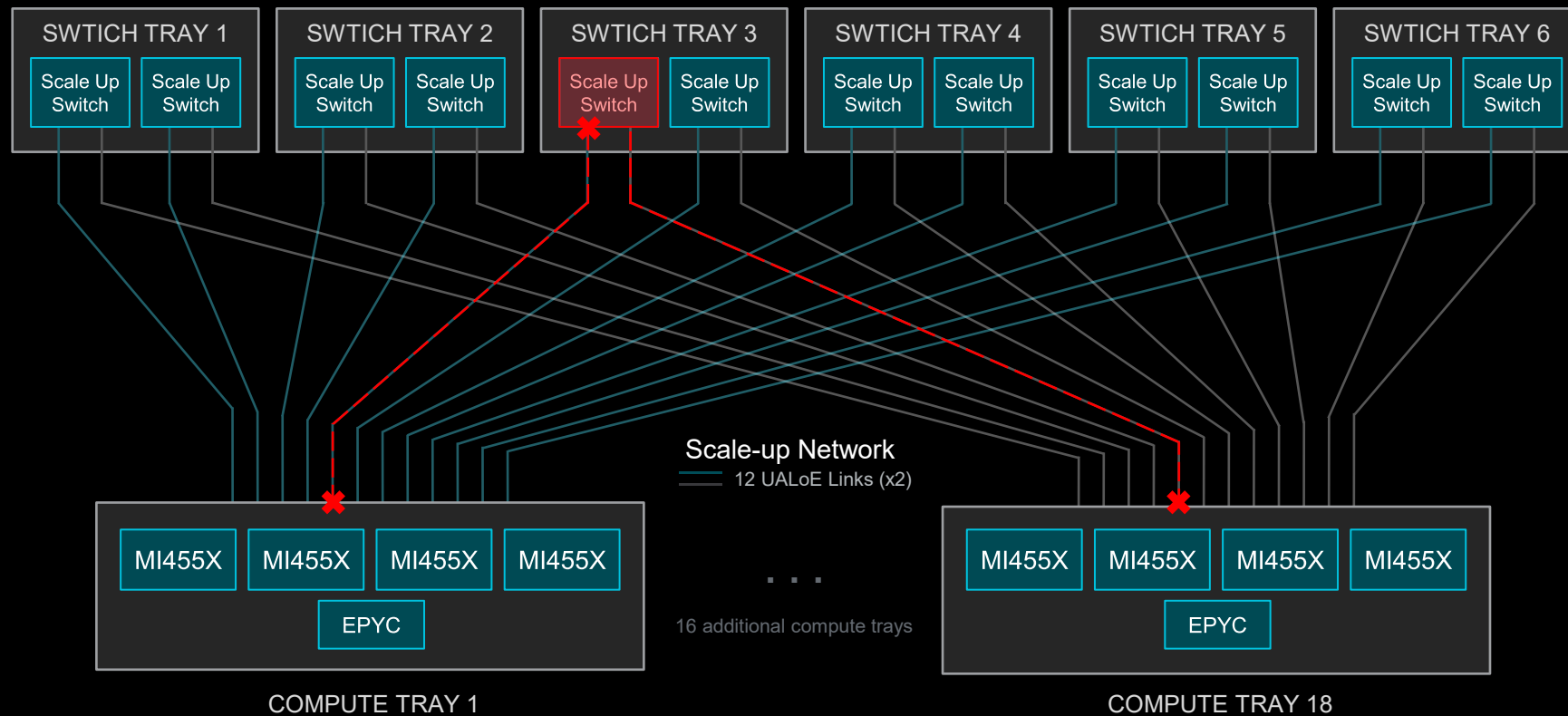
SINGLE LINK FAILURE

SINGLE SWITCH FAILURE

SWITCH TRAY FAILURE

1 of 12 switches lost | Impacts all VPods

DMA rebalances over 11 of 12 switches | WGP refs rebalance under SW control



Scale-up fault resilience

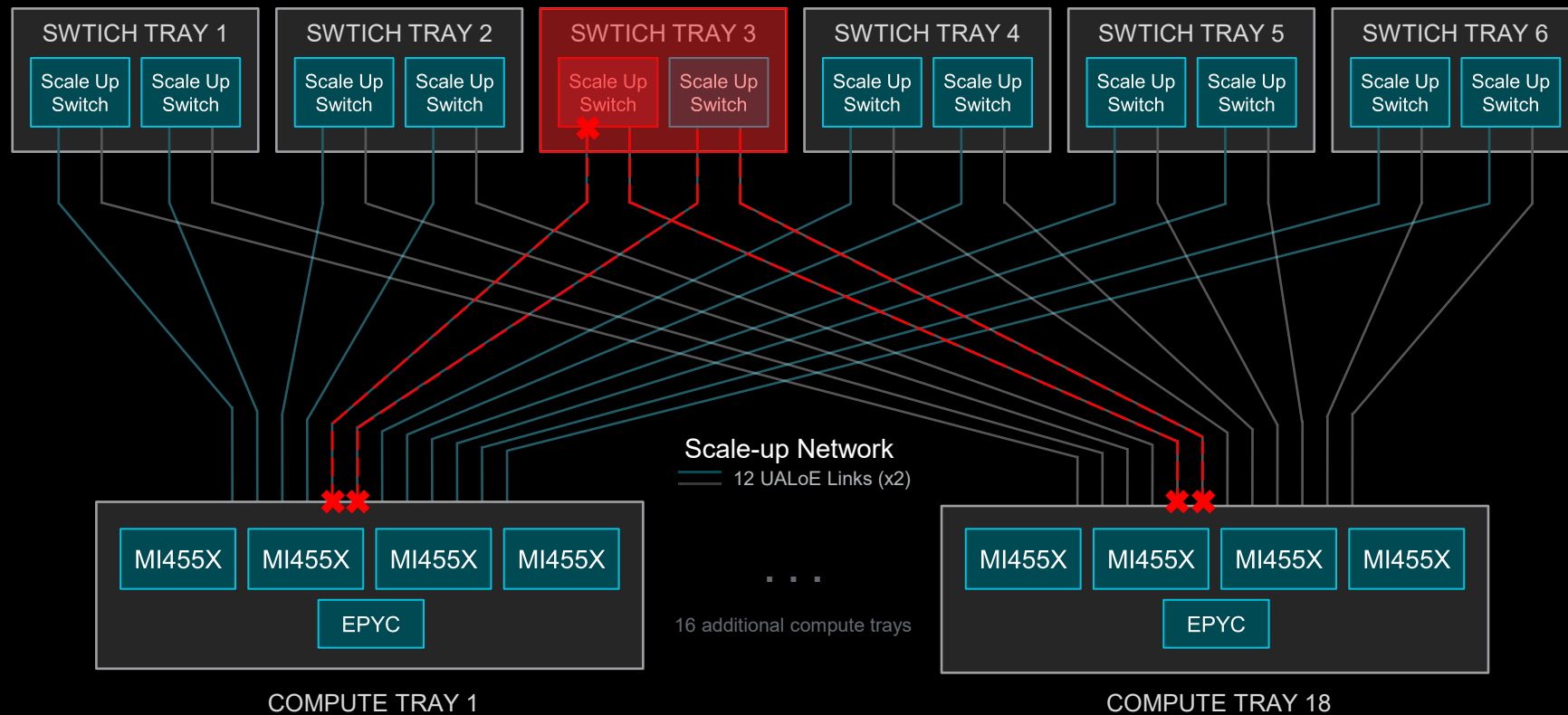
NORMAL OPERATION

SINGLE LINK FAILURE

SINGLE SWITCH FAILIRE

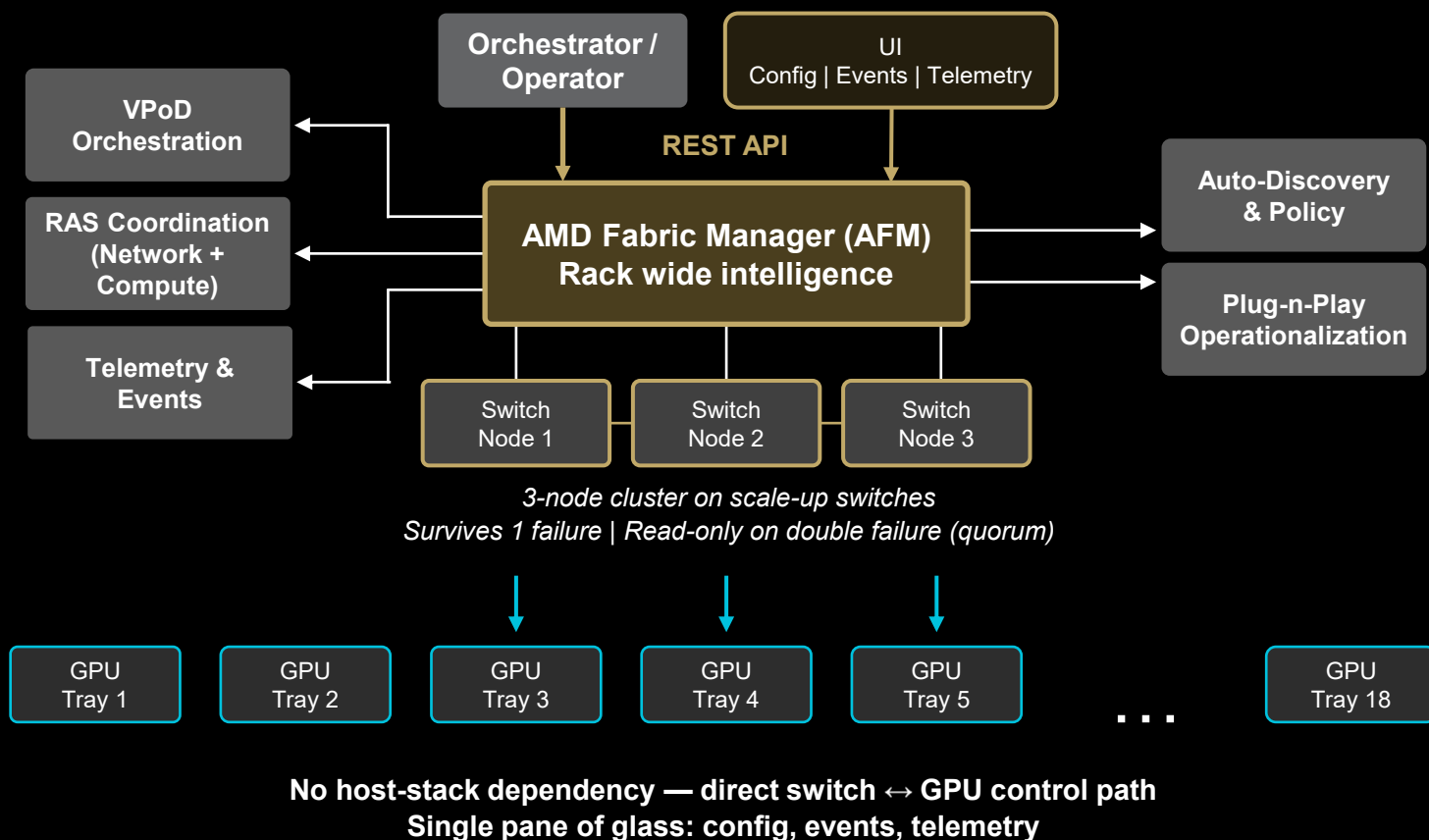
SWITCH TRAY FAILURE

Whole switch tray lost | Impacts all VPods
DMA rebalances over 5 of 6 trays | WGP refs rebalance under SW control

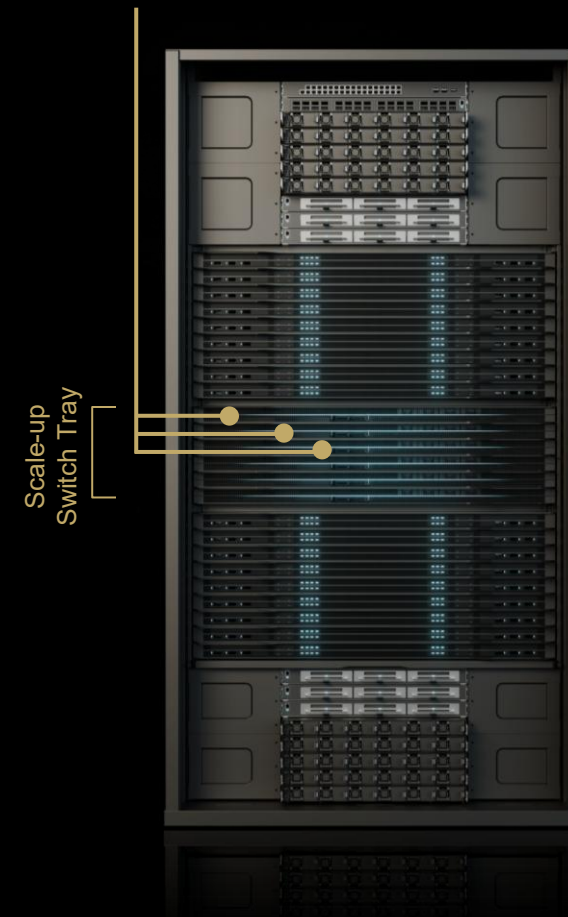


Rack-Wide Intelligence

AFM: Coherent, Resilient Control Plane



Fabric Manager (AFM) Manages Scale-up Network



AI NIC

Programmable NIC

AI NIC is designed to run programmable RDMA and network offloads to support low latency, high bandwidth storage and collective communications

Examples:

- Software Defined Networks (SDN)
- RDMA (RoCEv2, MRC, UEC and others)
- Security
- Storage (NVMeoF)

Deployed in multiple modes

①

Front-End Ethernet NIC

②

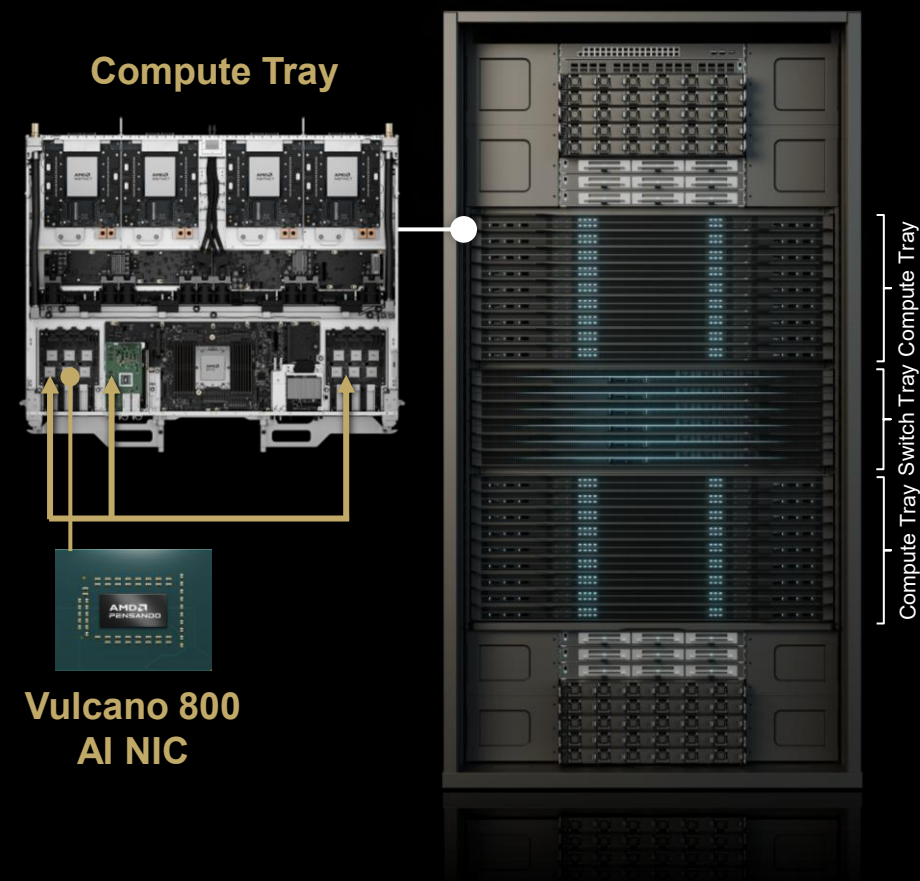
Scale-out RDMA NIC

③

Storage Initiator / Target NVMeoF

Multi-Generation Networking Innovation

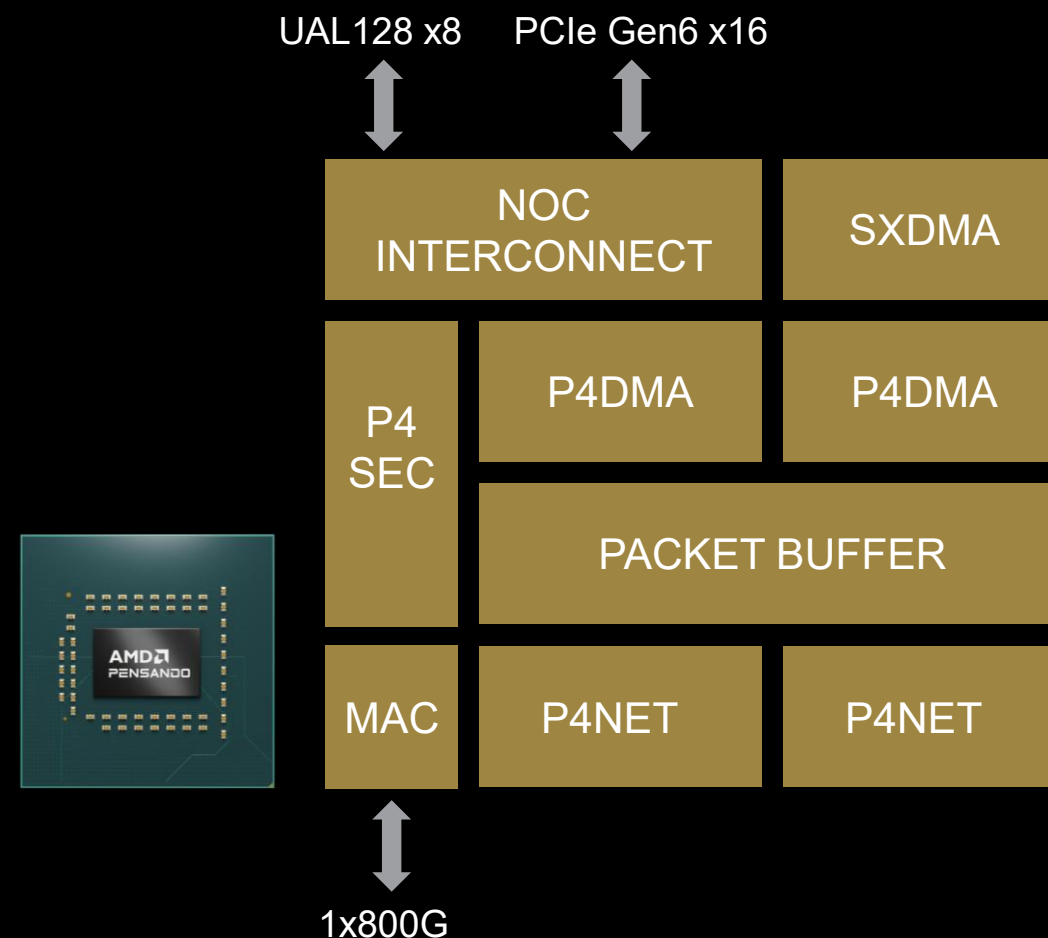
Helios Rack AI Solution



AMD Pensando™ Vulcano 800 AI NIC

Block Diagram

- P4-based architecture – **192** MPU
- ATS and RDMA translation services to P4DMA
- P4DMA programmability for transport (RoCEv2, MRC, UEC etc.)
- MPU added instructions to support congestion control
- Fine grained transaction support for relaxed ordering memory operations
- High-end frequency telemetry
- Low power and latency for the throughput



AMD Pensando™ Programmable Transport

Custom Protocols

Evolution of Network Transports Over Time

Mega Scale
Source Routing

Failure Handling, Resiliency
Multiplane, Path Probing (MRC)

Transport Efficiency
Multipathing, Selective ACKs (MRC, UEC)

Support for Lossy Fabrics
Window Based (MRC)

Legacy Transport
RoCEv2

AMD Pensando P4DMA Engine

Configurable Packet and Memory Parsing

Custom logic for implementing new protocols

Custom/New Collectives in Datapath

Adaptable telemetry to changing transports

AMD Pensando™ Programmable Congestion Control

Custom Protocols

Evolving Congestion Control Mechanisms and Required Framework

Bring Your Own CC
PCC Framework (UEC)

Newer Congestion Signals
UEC CSIG

Window Based Multipath Aware
MRC (Multipath Reliable Connection) NSCC, RCC

Classic Solutions
DCQCN (standard or enhanced)

AMD Pensando P4DMA Engine

Carry and Interpret Custom CC Signals

Maintain CC state for innovative algorithms

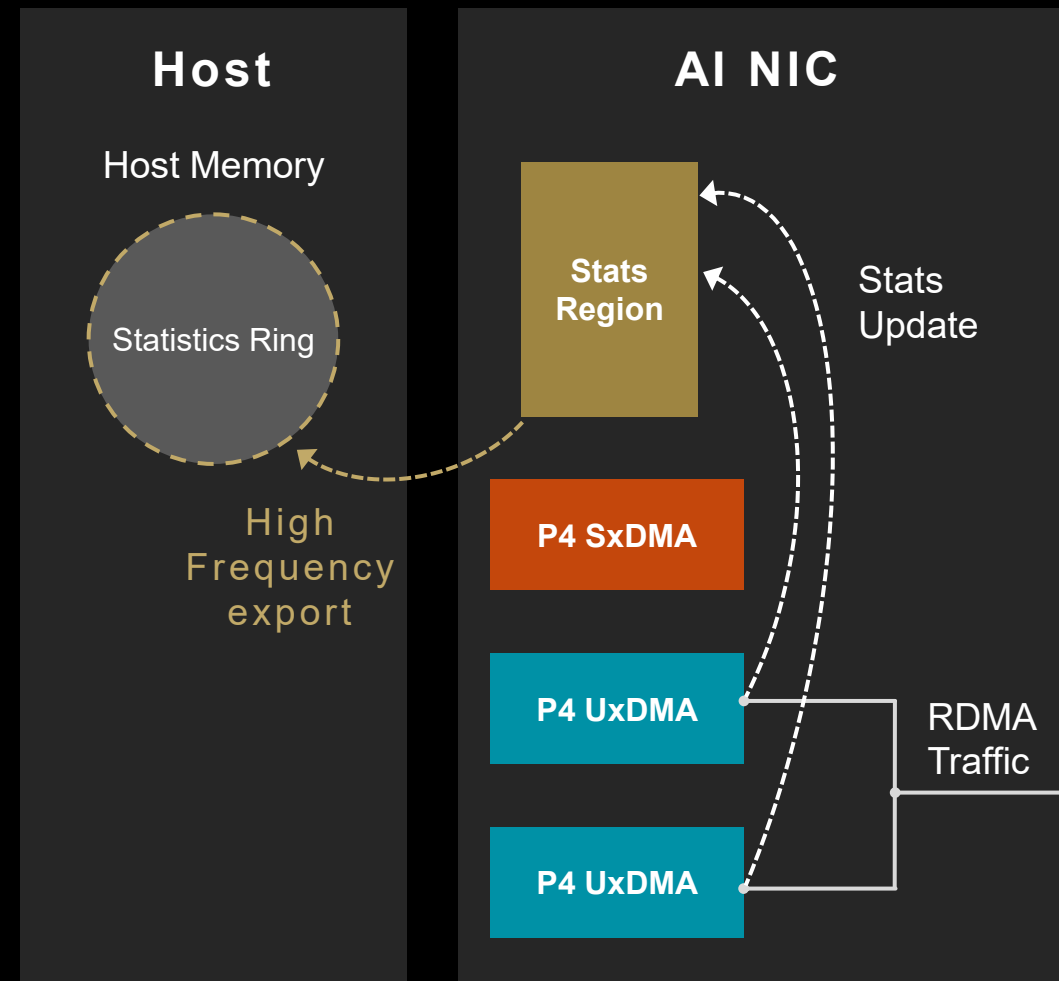
Adapt to high rate of CC events

Granular CC behavior measurement/analysis

AMD Pensando™ Vulcano 800

High Frequency Telemetry

- Programmable P4 pipeline to transfer telemetry from AI NIC to host memory at high frequency
- Provide multi-level telemetry – System, LIF, QP, Path etc.
- Offer comprehensive telemetry coverage, spanning transport and congestion control
- Capability to filter and select specific statistics for monitoring
- Flexibility to define the monitoring time window
- Support detailed, high-resolution monitoring



AMD Pensando™ Vulcano

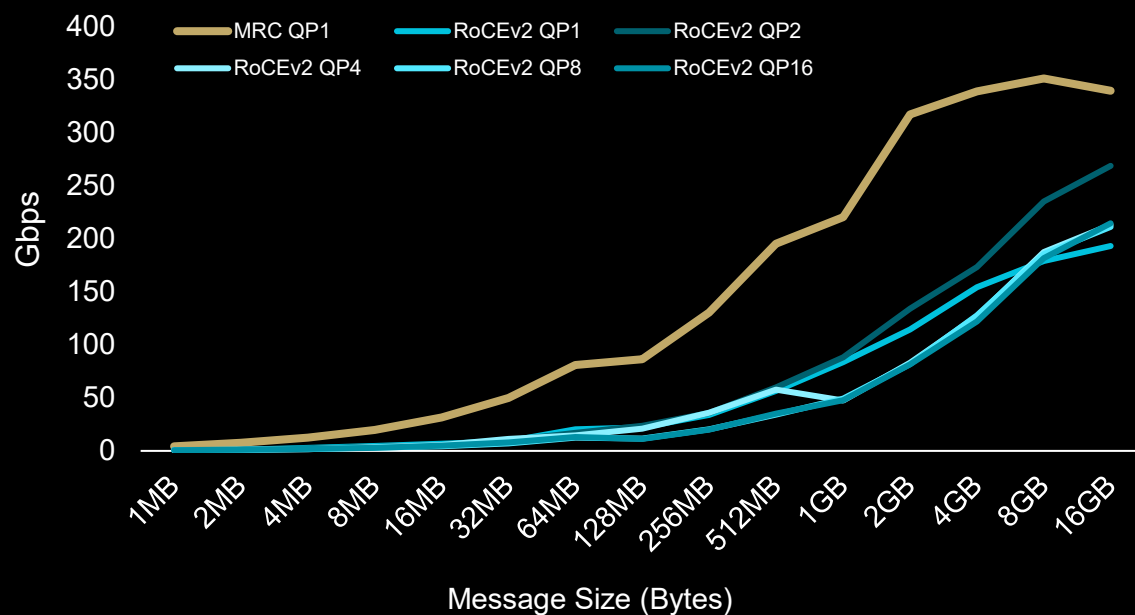
MRC Performance

Msg Size	Plane	1 – 4K QPs
>= 64KB	1x800G	800G Tx / 800G Rx
>= 64KB	2x400G	800G Tx / 800G Rx
>= 64KB	4x200G	800G Tx / 800G Rx
>= 64KB	8x100G	800G Tx / 800G Rx

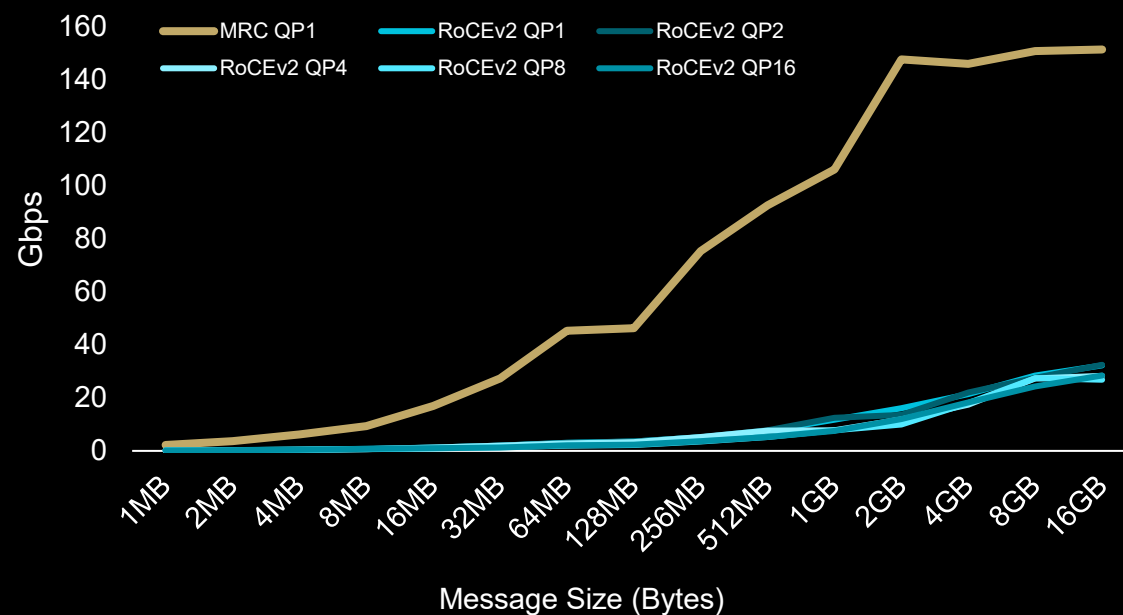
AMD Pensando™ Vulcano

MRC vs RoCEv2 Performance

All Reduce MRC QP1 vs RoCEv2 QP1, QP2, QP4, QP8, QP16



All Reduce 1% Drop MRC QP1 vs RoCEv2 QP1, QP2, QP4, QP8, QP16



**MRC performance
advantage over RoCEv2**

- Packet Spray
- Drop convergence using Selective Acknowledgement (SACK)

Summary

Massive growth in AI demand drives the need for datacenter-scale AI systems

AMD Helios rack co-designed with emerging AI workloads

- 72 GPUs operate as a unified compute engine
- ORW rack provides integrated power, cooling, and management

Scale-up network:

- Shared memory ld/st access to over 30TB of HBM4 at 1.8 TB/s/dir per GPU
- UALoE protocol provide flexible partitioning, security isolation, and multi-layer resiliency

Scale-out network:

- Pensando AI-NICs provide protocol flexibility, advanced load balancing & congestion control, and telemetry & error management

